

Efficient Simulation of Weak Nonlinearities in Continuous-Time Oversampling Converters

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Abstract—We propose a technique that enables the study of weak loop filter nonlinearities in a class of continuous-time delta-sigma modulators. The technique can easily be implemented in a tool intended to simulate discrete-time modulators with linear loop filters, thereby significantly reducing the simulation time. Thanks to this technique, the utility of the Schreier Delta-Sigma toolbox can be extended to include weakly nonlinear effects in the loop filter, thus enabling the architectural exploration of alternate loop filter and operational amplifier (opamp) topologies. The results from the use of our technique are compared to those obtained using a circuit simulator, and good agreement is seen.

Index Terms—Continuous-time (CT), delta, modulator, nonlinear, oversampling, quantization noise, sigma, Volterra, weakly.

I. INTRODUCTION

CONTINUOUS-TIME (CT) oversampling converters are becoming increasingly important in analog signal chains due to several attractive features like implicit antialiasing, resistive input impedance, and potential for low-power operation, as seen from several recently reported modulators [1]–[3]. The performance of these modulators is limited by thermal and $1/f$ noise, mismatch in the feedback digital-to-analog converter (DAC), nonlinearities in the loop filter, and clock jitter. Noise is easily addressed by impedance scaling. Techniques to mitigate nonlinearity caused by device mismatch in the feedback DAC have extensively been studied (see, for instance, [4] and the references therein). The modeling and analysis of clock jitter has also been addressed in the past [5]–[7]. However, only a few works deal with the effect of nonlinearities in the loop filter.

The CT delta-sigma modulator (CTDSM) design process typically proceeds as follows.

- 1) A loop filter transfer function and topology are chosen based on the desired in-band signal-to-quantization-noise ratio (SQNR) and signal transfer function (STF). A modulator is implemented using ideal integrators, and component values are derived after appropriate dynamic range scaling. Throughout this paper, we assume that active-RC integrators are used as the building blocks of the loop filter—this does not result in loss of generality.
- 2) After choosing an operational amplifier (opamp) architecture, the ideal integrators are replaced with real ones. The loop filter coefficients are tuned, if necessary, to account for finite-bandwidth effects of the opamps.

- 3) Transistor-level simulations of the modulator are run to ascertain the in-band signal-to-noise-plus-distortion ratio (SNDR) of the CTDSM. The obtained SNDR includes the effects of noise shaping of the quantization error, as well as its demodulation into the signal band due to nonlinearities in the loop filter.

- 4) If the integrators are not sufficiently linear, the SNDR obtained in step 3 above is much lower than that expected from linear analysis. To address this issue, the bias currents in the opamps are increased, and step 3 is repeated until specifications are met. Borrowing from experience with CT filters, it is generally accepted that loop filters synthesized using active-RC techniques are best suited for designing CTDSMs requiring high linearity.

Steps 1) and 2) can be accomplished using the simulation of a linear state space system in the discrete-time (DT) domain (using the principles of CT-to-DT equivalence). In the opinion of this author, the versatile Delta-Sigma toolbox is the most convenient way of doing this. Step 3) is time consuming, since the nonlinear differential equations describing the loop filter have to be solved, even if behavioral models are used for the opamps. To address this issue, several authors have developed special-purpose tools [8], [9] to simulate CTDSMs. Most circuit designers are not inclined to embark on such an exercise, and the alternative is to use a circuit simulator. Due to constraints on the simulation time, it is often not possible to thoroughly explore the design space of loop filter/opamp topologies and could lead to suboptimal designs.

Recognizing that the nonlinearity in the first integrator of the loop filter influences the performance of the modulator to the largest extent, some authors have quantified the effect of weak integrator nonlinearity in some special cases. The authors of [10] address the issue in a single-bit modulator employing a single-stage opamp-based active-RC integrator and return-to-zero (RZ) feedback DAC, concluding that nonlinearity results in harmonic distortion of the input sinusoid. Reference [11] investigates (using behavioral simulation) the relative significance of integrator nonlinearities in single-bit modulators with various loop filter topologies, concluding that the first integrator is the dominant contributor to distortion. In a recent paper [12], the effect of nonlinearities in the first integrator of a multibit CTDSM was analytically estimated for integrators based on single-stage and Miller-compensated opamps. The quantizer was considered as a source of additive noise, and simulations showed that expressions for the in-band SNR derived using this assumption compared well with those obtained using long circuit simulations.

From the above discussion, it is seen that the limited work on understanding and simulating nonlinear effects in CTDSMs is of two kinds. The first type [8], [9] includes works that describe

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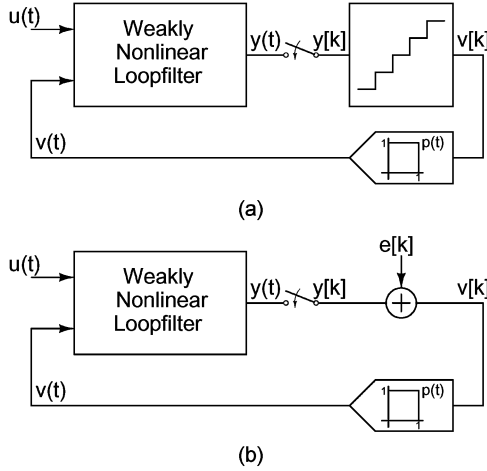


Fig. 1. CTDSM with (a) weakly nonlinear loop filter and (b) the quantizer replaced by an additive noise source.

specialized computer-aided design (CAD) tools—such an effort is not attractive to a circuit designer. In another body of work [10]–[12], simple and easy-to-use expressions with some insight are given, but their validity is restricted to a few special cases. Thus, a rapid way of simulating the effect of weak nonlinearities in the loop filter, without resorting to the development of a new CAD tool that solves the nonlinear integro-differential equations that describe the modulator, can be a useful aid to the designer to predict the performance and develop intuition about the effects of nonlinearity in CTDSMs.

The Volterra series approach is one compelling choice to analyze the effect of weak nonlinearities in systems. Applying this to a CTDSM as a whole does not seem to be justified due to the strongly nonlinear nature of the quantizer. However, if the quantizer is thought of as an additive noise source, the modulator becomes a mixed-domain (consisting of DT and CT signals) system with weak nonlinearity, with two excitations (the input to be digitized and quantization noise). This opens up the possibility of applying Volterra techniques for analysis and simulation. In this paper, we describe a way of accounting for weak nonlinearities in a CTDSM using a linear DT simulator. It is based on the method of current injection [13], originally intended to solve weakly nonlinear ordinary differential equations, and has been applied to the analysis of distortion in amplifiers and active filters (see [14] and [15], for example).

The proposed technique can easily be incorporated into the Delta-Sigma toolbox and results in a speedup of several orders of magnitude in simulation. The details of the method and its implementation form the subject of the rest of this paper, which is organized as follows. Section II develops the technique from first principles and describes the implementation. In Section III, we compare the results obtained from a circuit simulator to those from the proposed technique for several opamp and loop filter architectures. Conclusions are given in Section IV.

II. PROPOSED TECHNIQUE

The block diagram of a CTDSM is shown in Fig. 1(a). The sampling rate is 1 Hz—this does not lead to any loss of generality. $u(t)$ denotes the modulator input. The loop filter is weakly nonlinear, and its output is denoted by $y(t)$. The quantizer output

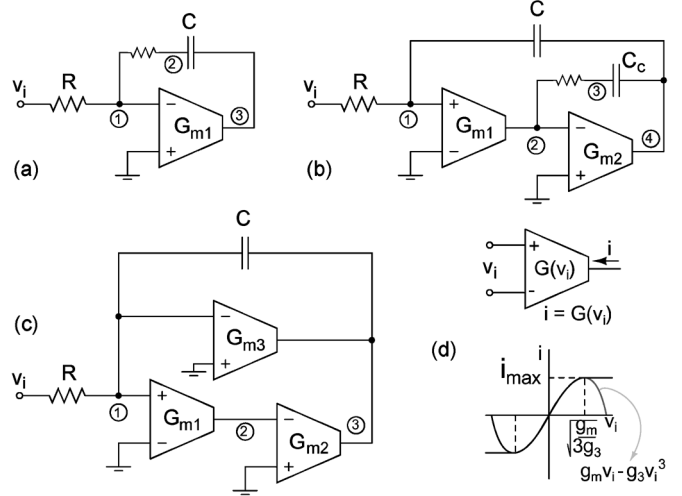


Fig. 2. Transconductor-based single-ended equivalent models for commonly used differential active- RC integrators based on (a) a single-stage operational transconductance amplifier, (b) a Miller-compensated two-stage opamp, and (c) a feedforward-compensated two-stage opamp. (d) Static characteristics of the transconductors, all of which are assumed to operate in their weakly nonlinear regions.

sequence $v[k]$ is fed back to the loop filter. $p(t)$ denotes the pulse shape of the feedback DAC. Making the usual additive noise approximation for the quantizer, we obtain the diagram in Fig. 1(b). In this figure, $e[k]$ is assumed to be a sequence independent of the quantizer input.

Throughout this paper, all circuits are assumed to be fully differential—hence, only third-order nonlinearities are considered. The loop filter is assumed to be synthesized using active- RC integrators of the types shown in Fig. 2(a)–(c), where single-ended equivalents of the differential circuits are shown. Parasitic capacitances at the input and output of each transconductor are not shown to reduce clutter in the diagram. The opamps consist of transconductors. As shown in Fig. 2(d), the static output current of every transconductor is assumed to be related to the input voltage as follows:

$$i = g_m v_i - g_3 v_i^3, \quad |v_i| \leq \sqrt{\frac{g_m}{3g_3}} \\ = \text{sign}(v_i) \frac{2}{3} g_m \sqrt{\frac{g_m}{3g_3}}, \quad |v_i| > \sqrt{\frac{g_m}{3g_3}}. \quad (1)$$

All transconductors are assumed to operate in their weakly nonlinear region of operation, i.e., for inputs where the magnitude of the linear term in (1) is much larger than the cubic term. We assume that only the transconductors are weakly nonlinear. This simplifies the equations but is otherwise not restrictive.

For the modulator in Fig. 1(b), the feedback DAC waveform is seen to be

$$v(t) = y_{\text{dac}}(t) + e(t)$$

where

$$y_{\text{dac}}(t) = \sum_k y[k] p(t - k) \\ e(t) = \sum_k e[k] p(t - k). \quad (2)$$

The dynamics of the modulator can be written as follows:

$$\mathbf{C}\dot{\mathbf{x}} + \mathbf{G}\mathbf{x} + \mathbf{G}_3\mathbf{x}^3 = [\mathbf{F}_1 \quad \mathbf{F}_2] [u(t) \quad v(t)]^T \quad (3)$$

where $\mathbf{x}$ is a column vector of node voltages in the loop filter, and $\mathbf{C}$, $\mathbf{G}$, $\mathbf{G}_3$, $\mathbf{F}_1$, and $\mathbf{F}_2$ are the usual capacitance, conductance, and input matrices arising from a modified nodal analysis (MNA) of the loop filter. $\mathbf{x}^3$ is a column vector of the cubes of the node voltages. If the loop filter has M nodes, $\mathbf{C}$, $\mathbf{G}$, and $\mathbf{G}_3$ are square matrices with size M . This special form of the MNA equations results because: 1) only the transconductors are nonlinear and 2) one of the transconductor input terminals is grounded (in the single-ended equivalent). If there are other nonlinearities (nonlinear capacitors, for example), additional terms would be needed in (3), but the discussion below would proceed along the same lines [13], [14].

To solve (2) and (3), we begin by writing $\mathbf{x}$ in its Volterra expanded form (bearing in mind the odd nonlinearity of the system and keeping only the first term in the expansion)

$$\mathbf{x}(t) \approx \mathbf{x}^{(1)}(t) + \mathbf{x}^{(3)}(t) \quad (4)$$

where $\mathbf{x}^{(1)}(t)$ and $\mathbf{x}^{(3)}(t)$ are the first- and third-order components of the Volterra series. We then proceed using the following “trick” (similar to the approach in [13])—we attempt to determine the states and output ($\hat{\mathbf{x}}$ and $\hat{y}$) if the inputs to the modulator [$u(t)$ and $e[k]$ in Fig. 1(b)] are scaled up by a factor z . $\hat{\mathbf{x}}$ can be related to $\mathbf{x}$ in the following manner. Since $\mathbf{x}^{(1)}(t)$ and $\mathbf{x}^{(3)}(t)$ are the first- and third-order terms of the Volterra expansion of $\mathbf{x}$, they should scale up by z and z^3 , respectively (when the inputs are scaled by z). Thus, $\hat{\mathbf{x}}$ is seen to be

$$\hat{\mathbf{x}}(t) \approx z\mathbf{x}^{(1)}(t) + z^3\mathbf{x}^{(3)}(t). \quad (5)$$

Since $\hat{y}_{\text{dac}}(t)$ is derived from $\hat{y}(t)$, which is a node in the loop filter, it is also of the form

$$\hat{y}_{\text{dac}}(t) \approx zy_{\text{dac}}^{(1)}(t) + z^3y_{\text{dac}}^{(3)}(t). \quad (6)$$

Using (5) and (6) in (3), we obtain

$$\begin{aligned} &\mathbf{C} \left(z\dot{\mathbf{x}}^{(1)} + z^3\dot{\mathbf{x}}^{(3)} \right) + \mathbf{G} \left(z\mathbf{x}^{(1)} + z^3\mathbf{x}^{(3)} \right) \\ &\quad + \mathbf{G}_3 \left(z\mathbf{x}^{(1)} + z^3\mathbf{x}^{(3)} \right)^3 \\ &= [\mathbf{F}_1 \quad \mathbf{F}_2] \begin{bmatrix} zu(t) & zy_{\text{dac}}^{(1)}(t) + z^3y_{\text{dac}}^{(3)}(t) + ze(t) \end{bmatrix}^T. \end{aligned} \quad (7)$$

Equating the coefficients of like powers of z in (7), we get

$$\mathbf{C}\dot{\mathbf{x}}^{(1)} + \mathbf{G}\mathbf{x}^{(1)} = [\mathbf{F}_1 \quad \mathbf{F}_2] \begin{bmatrix} u(t) & y_{\text{dac}}^{(1)}(t) + e(t) \end{bmatrix}^T \quad (8)$$

$$\mathbf{C}\dot{\mathbf{x}}^{(3)} + \mathbf{G}\mathbf{x}^{(3)} + \mathbf{G}_3 \left(\mathbf{x}^{(1)} \right)^3 = [\mathbf{F}_1 \quad \mathbf{F}_2] \begin{bmatrix} 0 & y_{\text{dac}}^{(3)}(t) \end{bmatrix}^T. \quad (9)$$

If one associates addition of $e[k]$ with quantization, (8) represents a modulator with a linear loop filter, as shown in Fig. 3(a). Equation (9) describes the system shown in Fig. 3(b), where the loop filter is still linear. The input and quantization noise are set to zero—the excitations to this linear system are currents that depend on the cubes of the node voltages of the system in (8).

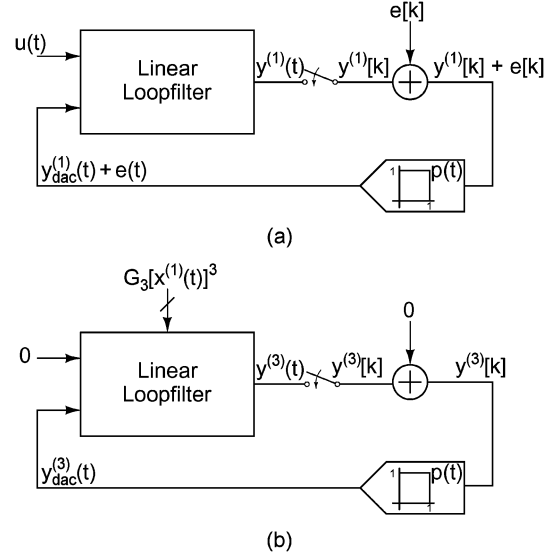


Fig. 3. Linear systems corresponding to (a) (8) and (b) (9).

The output of the modulator in Fig. 1(b) with the weakly nonlinear filter is seen to be

$$v[k] = y^{(1)}[k] + e[k] + y^{(3)}[k]. \quad (10)$$

This suggests the following method to determine $v[k]$, illustrated with a first-order CTDSM that uses an active- RC loop filter [Fig. 4(a)]. The “opamp” is a single-stage design, with a weakly nonlinear transconductor. The node voltages x_1 , x_2 , and x_3 are marked.

- 1) Determine the output sequence of the CTDSM with all nonlinear effects (in the loop filter) removed, i.e., by setting $g_3 = 0$. The resulting modulator is shown in Fig. 4(b). Let $\mathbf{x}^{(1)}(t)$ denote the vector of node voltages in the loop filter—which has a dimension of 3×1 in the example. Denote the output sequence of the modulator by $v^{(1)}[k]$.
- 2) Create a “modulator” loop with the linear part of the loop filter but with the quantizer bypassed, as shown in Fig. 4(c). The main input to the loop filter [corresponding to $u(t)$ in Fig. 4(b)] is set to zero, and a current $g_3[x_1^{(1)}]^3$ is injected in parallel with the nonlinear current source of the original modulator. $x_1^{(1)}$ is known from step 1 above. If the only nonlinear elements in the loop filter are transconductors, the number of currents to be injected is equal to the number of transconductors. Denote the output sequence of this linear system as $v^{(3)}[k]$.
- 3) Compute the power spectral density (PSD) of $v^{(1)}[k] + v^{(3)}[k]$ to estimate the in-band SNR of the modulator—this now accounts for quantization noise, as well as weak nonlinearities in the loop filter.

From the discussion above, it is seen that the problem of calculating the effects of weak nonlinearity in the loop filter can be reduced to one of simulating two systems, both of which have CT linear loop filters. The first system [Fig. 4(a)] has a CT linear loop filter and a quantizer, while the second [Fig. 4(b)] is simply a linear system comprising of the same filter (but without the quantizer), whose inputs are derived from internal signals in the first system. The difference between the true modulator (with

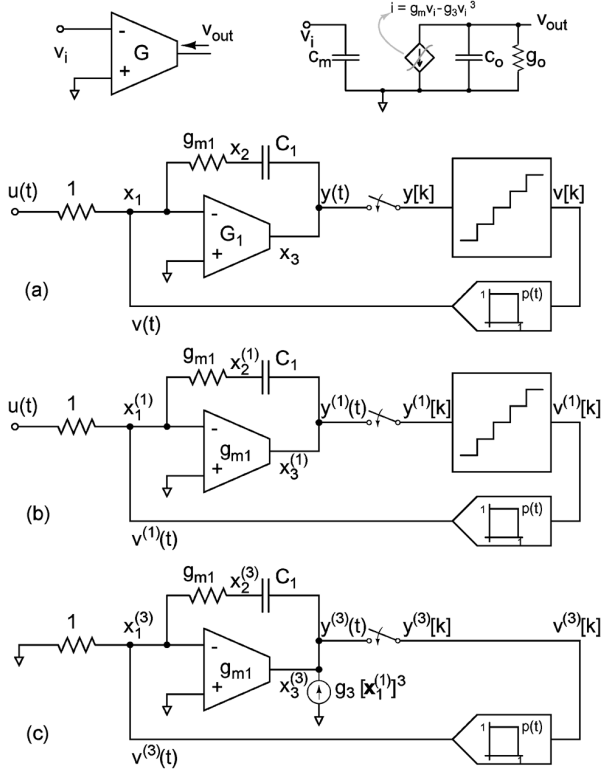


Fig. 4. Approximate analysis of weak nonlinearity in a first-order CTDSM. (a) Modulator with nonlinear loop filter. (b) Step 1: solving the linear modulator. (c) Step 2: using the results of step 1 to inject a current into a linear modulator without the quantizer. The in-band SNDR is computed from the PSD of $v^{(1)}[k] + v^{(3)}[k]$.

filter nonlinearity) and our approximation is given as follows. The real input to the quantizer is a function of the (weak) nonlinearities in the loop filter, while that in our approximation does not depend on these nonlinearities. Intuitively, it is expected that the characteristics of quantization noise are not very different in the two cases. Furthermore, any errors in the approximation are suppressed in the signal band due to the high gain of the loop filter. In our approach, effects like overload and maximum stable amplitude are modeled (thanks to the quantizer in the first system) but do not account for weak loop filter nonlinearity. This is not relevant in practice, since quantizer saturation is a much stronger phenomenon, and predictions of instability and maximum stable amplitude are largely unaffected by weak nonlinearities in the loop filter. This is confirmed by the simulation results given in Section III.

The mechanism of degradation of the in-band SNDR due to weak nonlinearity can be understood as follows. Referring to Fig. 4, the output is seen to be $v^{(1)}[k] + v^{(3)}[k]$. The in-band noise of $v^{(1)}[k]$ corresponds to that of a linear modulator, while the in-band components of $v^{(3)}[k]$ depend on $\mathbf{x}^{(3)}$. This, in turn, depends on $[\mathbf{x}^{(1)}]^3$, which are *cubed versions* of linear combinations of (filtered) input and quantization noise. Thus, the components of $\mathbf{x}^{(1)}$ at frequencies outside the signal band intermodulate and fold into low frequencies, reducing the SNDR.

The systems in Fig. 3(a) and (b) involve CT filters. If the systems can be simulated in DT, one could take advantage of a large body of knowledge developed for simulating DT modulators [16]. A method of accomplishing this is given in the

Appendix—it exploits the principle of the CT-to-DT equivalence in a manner that enables the effects of nonlinearity to be included.

III. SIMULATION RESULTS AND DISCUSSION

In this section, we present results obtained using our technique (implemented as given in the Appendix) and compare them with the results from a circuit simulator. Results for all possible opamp, loop filter, and DAC pulse shapes and modulator topologies cannot be shown here—we concentrate on single-loop modulators with non-RZ (NRZ) and RZ feedback DACs based on the cascade of integrators with feedforward (CIFF) and cascade of integrators with feedback (CIFB) topologies. Two-stage opamps based on feedforward and Miller compensation are used in our designs. The achievable speedup in simulation is reflected by the following. The time required to generate a 4096-point data record was 1.1 s using our technique, while a circuit simulator took 380 s (for $\text{reltol} = 10^{-6}$ and a maximum step size of 1 ms).

A. Multibit Modulators

Fig. 5(a) and (b) shows the impedance and sampling rate normalized schematics of a third-order nine-level CTDSMs, where the loop filters are implemented using the CIFF and CIFB architectures. The actual signal bandwidth and sampling rate are 24 kHz and 6.144 MHz, respectively, resulting in an oversampling ratio of 128. The loops are designed to have a maximally flat noise transfer function (NTF) with an out-of-band gain (OBG) of 2 when ideal opamps are used. The DACs are of the switched-current variety, with an NRZ pulse shape. As indicated in the Appendix, the quantizer step size is 2.

The opamps are two-stage feedforward-compensated designs, with a dc gain of about 54 dB when loaded by the next integrator. The parasitic capacitances at the input, intermediate, and output nodes are indicated in Fig. 5(c). An ideal summing amplifier is used in the CIFF modulator.

Fig. 6 compares the magnitudes of the NTF when ideal and real opamps are used. Observe that the NTF matches the ideal response at in-band frequencies, and the finite opamp bandwidth results in the peaking of the NTF at out-of-band frequencies.

Fig. 7(a) and (b) shows the in-band PSDs for the CIFF and CIFB modulators in Fig. 5(a) and (b), respectively. There are three curves in each figure—the PSD obtained using our method, that obtained by simulating a behavioral model of the CTDSM in a commercial circuit simulator (Spectre), and that of a modulator with a linear loop filter (including finite-bandwidth effects but with all nonlinear effects turned off). In the simulations, the input to the modulators was a 6-kHz input sinusoid with an amplitude of 75% of the quantizer range (-2.5 dBFS). Excellent agreement is seen between our technique and the circuit simulation.

1) *Performance With Varying Input Amplitude:* Fig. 8 shows the in-band SNDR of the CIFF modulator in Fig. 5(a) as the input amplitude is varied from -40 to 0 dBFS. We see that the proposed technique does a good job of predicting the performance of the modulator when weak nonlinearities are considered. The SNDR of an ideal modulator is also shown for comparison. Since our method incorporates the quantizer (see

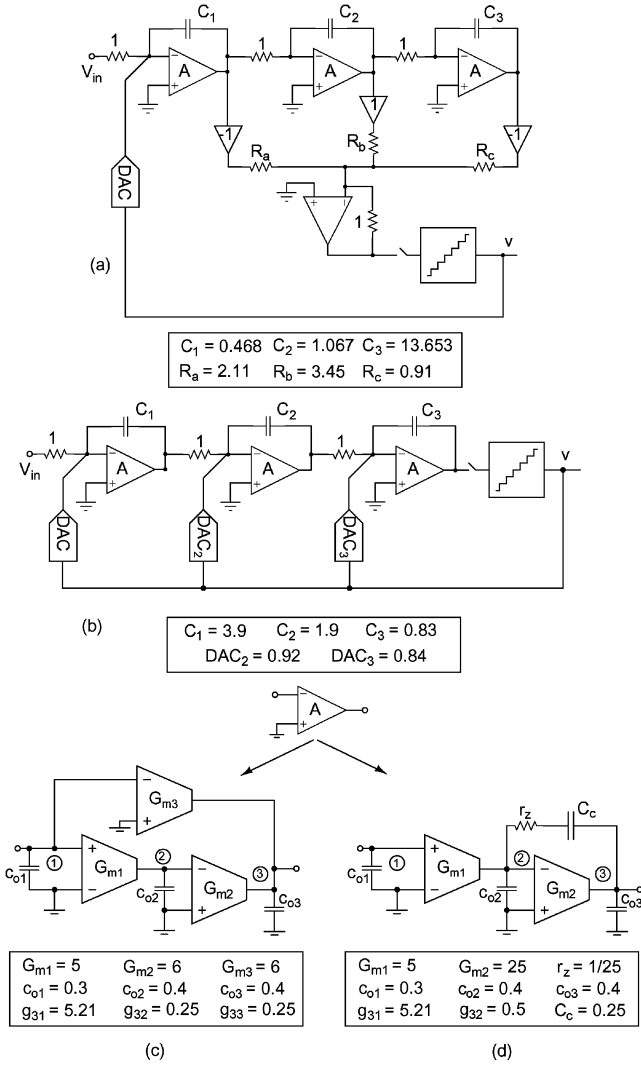


Fig. 5. Nine-level third-order CTDSMs with a switched-current feedback DAC. (a) With a CIFF loop filter. (b) With a CIFB loop filter. (c) Schematic of the feedforward-compensated opamp used. (d) Schematic of the Miller-compensated opamp used. The component values correspond to modulators whose sampling rate is normalized to 1 Hz, and impedances are scaled so that the input resistors of the integrators are 1Ω .

Fig. 4), effects related to quantizer behavior (like instability due to overloading) are quite accurately predicted. This makes sense, since weak nonlinearities in the loop filter cannot be expected to be influential when the quantizer saturates.

2) *Intermodulation Distortion*: To test the robustness of our technique with out-of-band signals, the CIFF modulator in Fig. 5(a) was excited by an input comprising of three tones—a 6-kHz sine wave with an amplitude of 1 V and “interfering” tones at 684 kHz and 1.344 MHz, both having an amplitude denoted by A_i . One of the third-order intermodulation products of the out-of-band tones occurs at 24 kHz. It is well known that a modulator with a CIFF loop filter has a peak in the STF. Thus, the feedback DAC injects amplified versions of the interferers into the loop filter. The component values of the design in Fig. 5(a) were derived after coefficient scaling considering only in-band signals—subjecting the modulator to strong interferers resulted in slewing and instability. This was addressed by reducing the parasitic capacitors in the opamp by a factor of 2 and increasing the linear ranges of G_{m1} and G_{m2} by factors

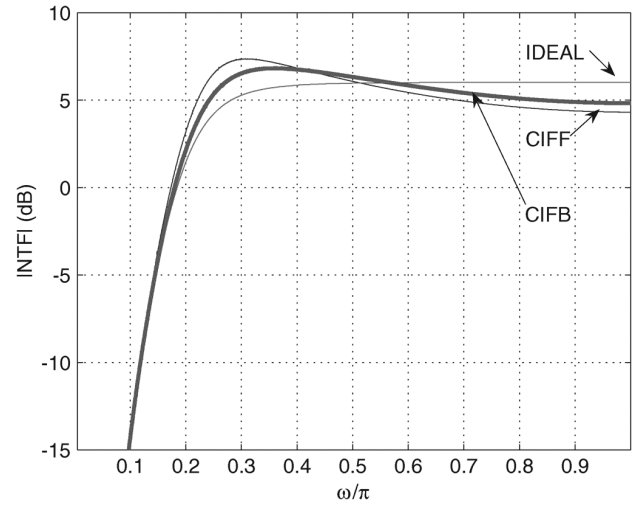


Fig. 6. NTF of the CIFF and CIFB modulators with ideal and real opamps.

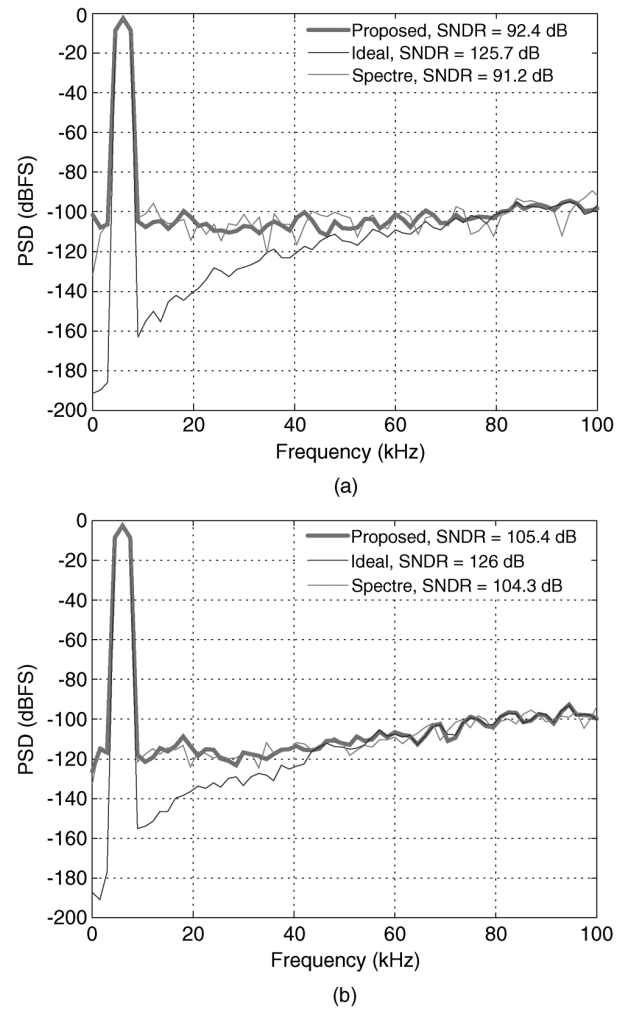


Fig. 7. Comparison of the PSDs of CTDSMs using the proposed technique with those obtained from the circuit simulation for designs using (a) a CIFF loop filter and (b) a CIFB loop filter. In both cases, the PSD for a modulator with a linear loop filter is also shown. The input is a -2.5 -dBFS tone at 6 kHz, and $f_s = 6.144$ MHz. A switched-current NRZ feedback DAC and feedforward-compensated opamps are used.

of $2\sqrt{2}$ and 2, respectively—thereby allowing the modulator to operate without slewing, with A_i as large as 1.4 V. Apart from

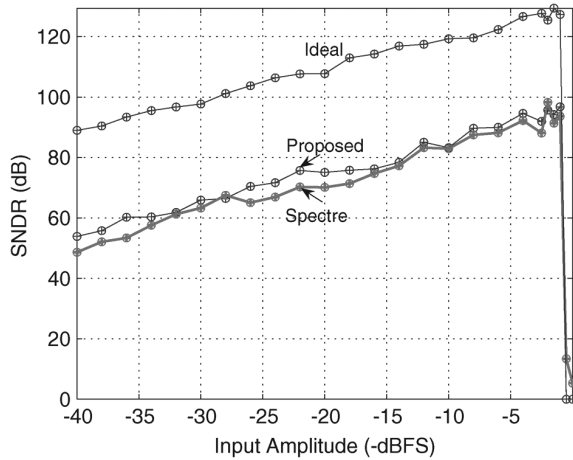


Fig. 8. Comparison of the SNDR obtained using the proposed technique with that from the circuit simulation. The ideal SNDR is also shown for reference. The modulator is that in Fig. 5(a), with feedforward-compensated opamps.

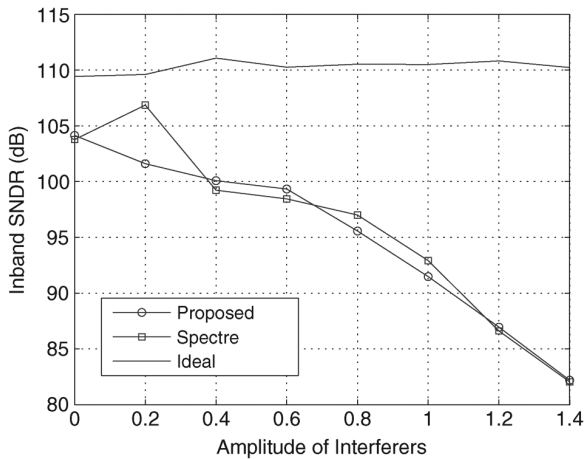


Fig. 9. Comparison of the in-band SNDR due to interfering out-of-band tones at 684 kHz and 1.344 MHz. The ideal SNDR is also shown for reference. The modulator is that in Fig. 5(a), with feedforward-compensated opamps.

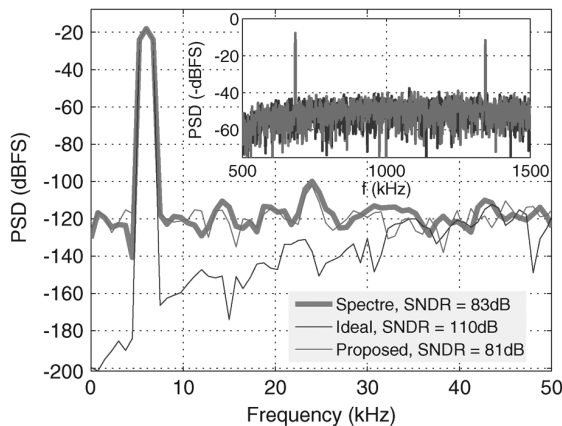
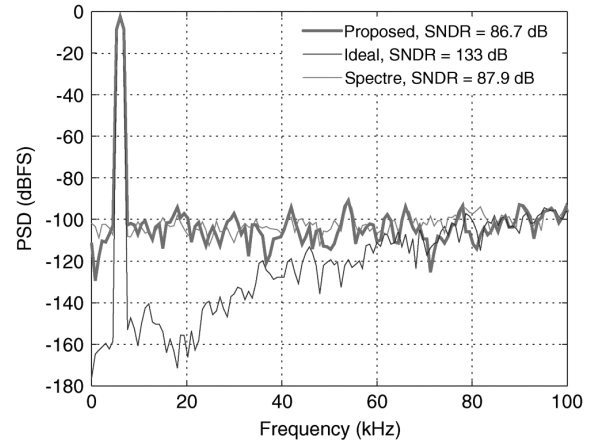
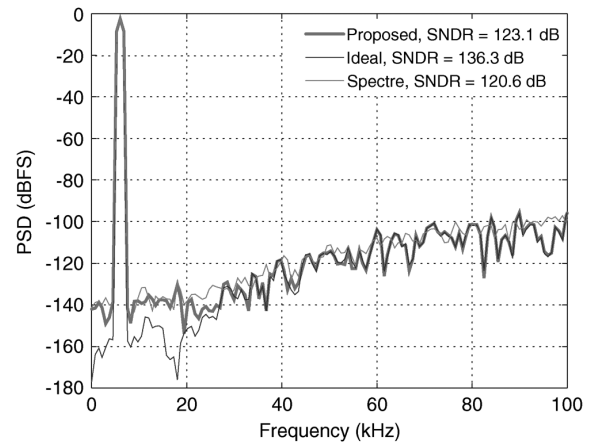


Fig. 10. PSD due to interfering out-of-band tones at 684 kHz and 1.344 MHz. The ideal PSD is also shown for reference. The noise floor is increased, and an intermodulation tone at 24 kHz is seen. The inset shows the spectrum around the two interfering tones.

introducing a distortion component at 24 kHz, the interferers also raise the in-band noise floor. Fig. 9 shows the in-band SNDR as a function of the amplitude (A_i) of the out-of-band



(a)



(b)

Fig. 11. Comparison of the PSDs of Miller-compensated-opamp-based CTDSMs using the proposed technique with those obtained from the circuit simulation for nine-level modulators using (a) a CIFF loop filter and (b) a CIFB loop filter. In both cases, the PSD for a modulator with a linear loop filter is also shown. The input is a -2.5 -dBFS tone at 6 kHz, and $f_s = 6.144$ MHz. A switched-current NRZ feedback DAC is used.

tones. Good agreement is seen between the SNDRs obtained from circuit simulation and those from our technique. As expected, the SNDR of a linear modulator remains unchanged in the presence of interferers. Fig. 10 shows the modulator output PSD when $A_i = 1.4$ V. The inset shows the spectrum around the interferer frequencies. The apparent discrepancy (of 2 dB) in the predicted SNDR between Figs. 9 and 10 stems from the difference in the lengths of the data records used to generate the two figures (4096 versus 8192 points).

Other results are not given here due to space limitations, but simulations show (as expected) that weak nonlinearities on the second and third integrators do not contribute to degradation in the in-band SNR in the CIFF case.

3) Miller-Compensated Opamps: Fig. 11 compares the performance of CIFF and CIFB modulators synthesized using Miller-compensated opamps. The details of the opamp are given in Fig. 5(d). The loop filter is modified by adding resistive feedback around the second and third integrators so that the NTF has complex zeros. Apart from the good agreement between the proposed technique and simulation, observe that the CIFB design outperforms the CIFF modulator by about 30 dB.

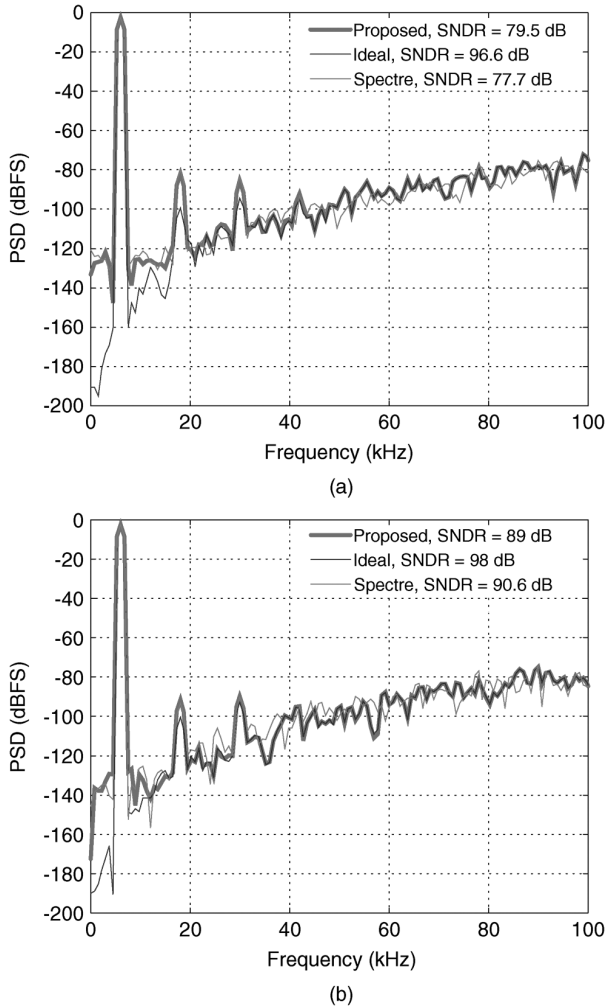


Fig. 12. Comparison of the PSDs of third-order three-level modulator outputs for a -2.5 -dBFS input tone at 6 kHz. $f_s = 6.144$ MHz when (a) a CIFF loop filter and (b) a CIFB loop filter is used. The DAC is of the switched-current NRZ feedback type.

4) *Number of Quantizer Levels*: Fig. 12 compares the in-band PSDs computed using our technique with those obtained from a circuit simulator for three-level CIFF and CIFB modulators. Feedforward-compensated opamps and current-steering NRZ DACs were used. Due to stability concerns, the NTF was chosen to have an OBG = 1.5. It is seen that the proposed method does a good job of predicting the in-band SNDR due to loop filter nonlinearities.

5) *DAC Pulse Shape (RZ DAC)*: To demonstrate the efficacy of our technique with different DAC pulse shapes, the CIFF modulator in Fig. 5(a) was simulated with an RZ feedback DAC. The NTF was modified to have complex zeros in the signal band. Feedforward opamps of the kind shown in Fig. 5(c) are used in the loop filter. Using an opamp with the same linearity as that employed in the NRZ DAC case resulted in the slewing of the first opamp. This makes sense, since the RZ feedback waveform consists of steps that need to start from zero every time. One rule of thumb followed in CTDSM design is to avoid the slewing and clipping of integrator outputs to avoid a severe drop in the in-band SNR [17]. To prevent slewing, the opamps needed to be made *much* more linear— $g_{31} = 0.163$, $g_{32} = 0.0625$,

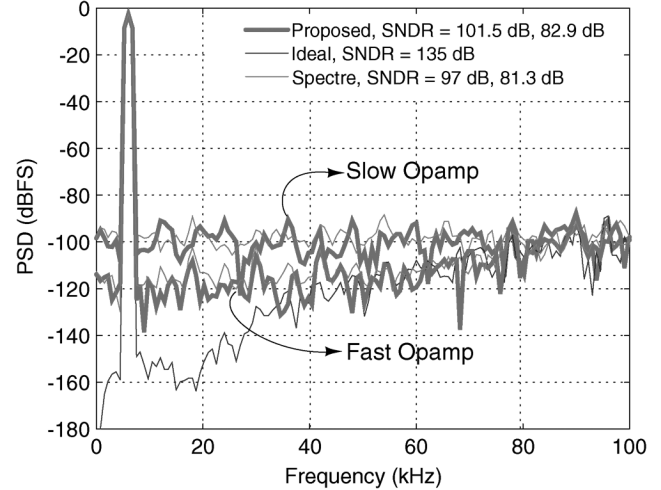


Fig. 13. Comparison of the PSDs of a CIFF modulator using an RZ DAC. The input is a -2.5 -dBFS input tone at 6 kHz. $f_s = 6.144$ MHz. The curves with the higher SNDR correspond to a feedforward-compensated opamp with the same bandwidth as that in Fig. 5(c), while those with a lower SNDR are when the opamp bandwidth is reduced by doubling all its parasitic capacitances.

and $g_{33} = 0.0625$. Simulations were also run by deliberately slowing down the opamp by doubling c_{o1} , c_{o2} , and c_{o3} . Fig. 13 shows the PSDs obtained by using our technique and a circuit simulator. The ideal PSD is also given for comparison. As expected, the modulator SNR is degraded when a slower opamp is used. When compared with the performance of modulators based on NRZ DACs, it is seen that the RZ design poorly performs in spite of using a far more linear opamp. Although RZ DACs do not cause distortion due to rise/fall-time asymmetry, they greatly increase the requirements on the linearity of the loop filter.

B. Single-Bit CTDSMs

The additive quantization noise assumption was the basis of our simplified technique to estimate the effect of weak loop filter nonlinearities on the performance of CTDSMs. This is not a very good assumption in single-bit modulators—so one should expect discrepancies between the SNR predicted by our technique and that by the circuit simulation. Fig. 14 shows the low-frequency PSDs of CIFF and CIFB third-order single-bit modulators. The NTF has an OBG = 1.5, with complex zeros optimally spread in the signal band. NRZ current-steering DACs are used in both modulators. The opamps are feedforward compensated with small-signal (trans)conductances and capacitances, as indicated in Fig. 5(c). The nonlinear coefficients of the transconductors used were $g_{31} = 62.5$, $g_{32} = 4$, and $g_{33} = 4$, respectively. From the simulation results, it is seen that loop filter nonlinearity in a single-bit modulator largely appears as distortion, and somewhat surprisingly, the predictions of the Volterra-based approach are close to those from circuit simulation. It appears as if the errors made in approximating the quantizer by an independent noise source are shaped out of the signal band by the modulator.

As a final illustration of the accuracy of our technique, we show the results of a blocker test in the CIFB design. The modulator input consists of a -20 -dBFS tone at 6 kHz and two blockers at 684 kHz and 1.344 MHz with an amplitude

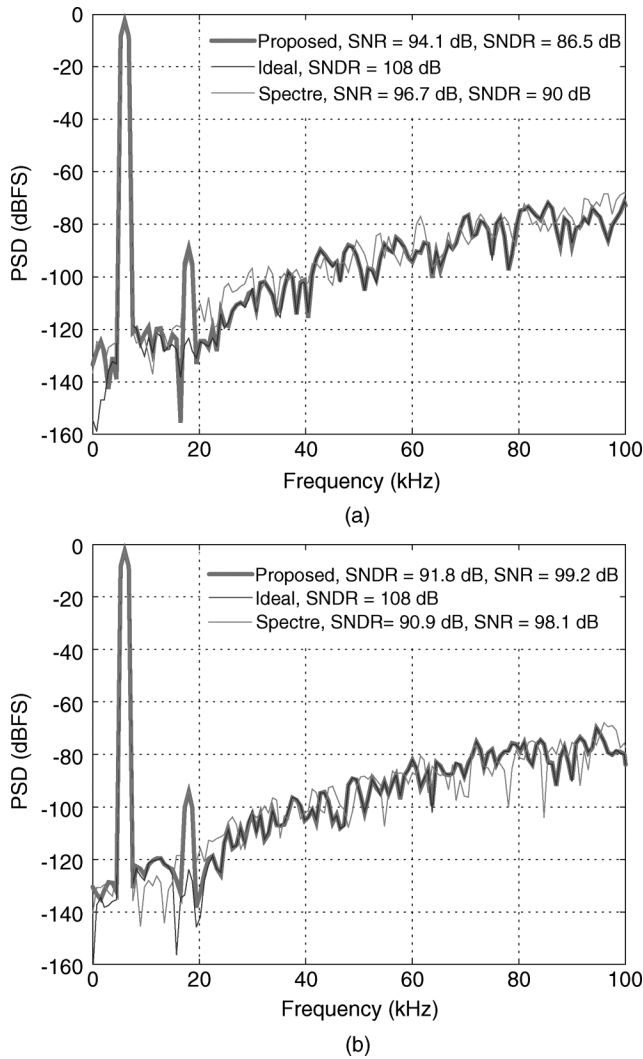


Fig. 14. Comparison of the in-band PSDs for single-bit modulators with (a) a CIFF loop filter and (b) a CIFB loop filter. An NRZ switched-current feedback DAC is used. The NTF has an OBG = 1.5, with complex zeros optimally spread in the signal band. The input is a -2.5 -dBFS tone at 6 kHz.

of -12 dBFS. Fig. 15 compares the ideal PSD with those obtained from circuit simulation and our technique. Good agreement is seen. The inset shows the PSD between 500 kHz and 1.5 MHz—thanks to the CIFB architecture, the 684 kHz tone is attenuated by 8 dB, while the 1.344-MHz tone is masked out by the shaped quantization noise.

One aspect of the simulation results presented in this paper that deserves some attention is the relative performance of modulators employing CIFF and CIFB loop filters. Note that, in all our simulations, CIFB modulators outperform their CIFF counterparts—which might be surprising, given the general belief in the design community that CIFF loop filters tend to be more linear than their CIFB counterparts. The intuition behind the reduced distortion of the CIFB modulators is the following. The unity gain frequency of the first integrator in the CIFB loop is much lower than that of the first integrator of the CIFF design (by a factor of 8, as seen in Fig. 5). Thus, *if the same opamp* is used in both cases, the loop gain around the first opamp is higher in the CIFB case—resulting in a reduced swing at the “virtual

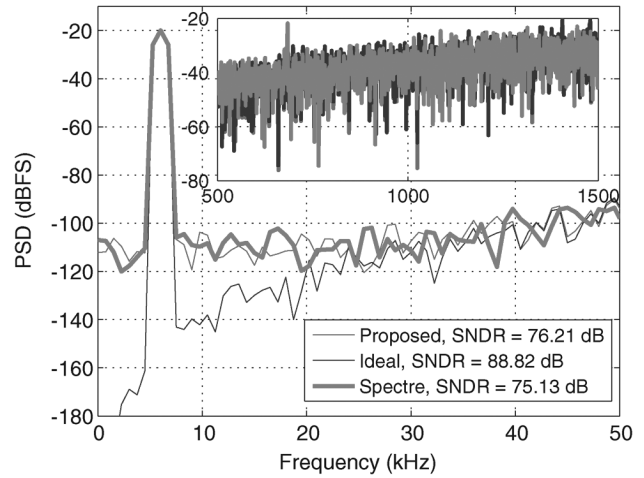


Fig. 15. PSD comparison for a CIFB single-bit modulator using an NRZ switched-current feedback DAC—the input consists of a -20 -dBFS tone at 6 kHz and two -12.5 -dBFS tones at 684 kHz and 1.344 kHz, respectively.

ground” of the opamp. Since distortion occurs due to currents proportional to the cubes of the node voltages (Figs. 3 and 4), we see that the CIFB designs have improved linearity.

IV. CONCLUSION

We have presented a way that allows a designer to rapidly predict the effect of weak loop filter nonlinearities in CT oversampled converters. The proposed technique is an adaptation of the method of current injection (intended for weakly nonlinear CT systems) to a mixed-domain system consisting of a quantizer embedded in a feedback loop. We have shown that weak nonlinearities can be analyzed by simulating two systems with linear loop filters and presented a method of running these simulations in DT. It thus becomes possible to take advantage of the large body of work in the area of simulating DT $\Delta\Sigma$ modulators (particularly [16]). The results obtained from the proposed technique agreed well with those from circuit simulation for several opamp, DAC, and modulator topologies. The techniques presented in this paper can be used to investigate other effects like the nonlinearity of the integrating capacitors and the nonlinear output impedance of the transconductors. It also enables rapid architectural exploration of various CTDSM and opamp topologies (if implemented as suggested in the Appendix) without the effort of developing a specialized CAD tool for this purpose.

APPENDIX

SIMULATING WEAK NONLINEARITY IN CTDSMS IN DISCRETE TIME

The simulation of CTDSMs in DT has traditionally been handled [5] by assuming that the modulator input is slowly varying, so that it can be approximated by a zero-order-held waveform at the sampling rate. This enables the “mimicking” of the behavior of the CT loop filter by a sampled-data system—and the modulator can rapidly be simulated in DT. In the discussion that follows, we describe a simple technique of using the popular Delta-Sigma toolbox (or any engine intended to design and simulate DT oversampled converters with *linear* loop filters) to

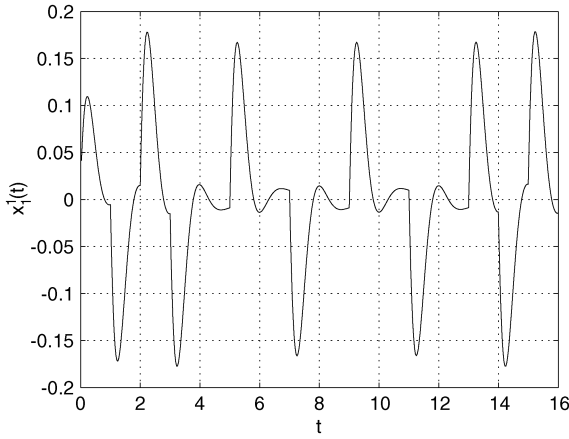


Fig. 16. Example waveform at the “virtual ground” node of the first opamp in a CTDSM using an active-RC integrator. The modulator is operating at a sampling rate of 1 Hz.

estimate the effect of weak *nonlinearities* of a CTDSM, thereby extending the capabilities of the toolbox.

From the discussion in Section II, we see that all the cubes of node voltages of the loop filter ($[x^{(1)}]^3$) are needed to simulate the effect of nonlinearity. Fig. 16 shows the voltage at the virtual ground node in a CTDSM that uses a feedforward-amplifier-based integrator. Clearly, these (and other) nodal voltages cannot be approximated well by zero-order-held waveforms at the CTDSM sampling rate. Hence, the loop filter needs to simulate at a higher rate P , as explained with the help of Fig. 17. Fig. 17(a) shows the input $u(t)$ and the DAC output $v(t)$. $u(t)$ can be approximated by $\hat{u}(t)$, which is zero-order held at P Hz. $v(t)$ changes every 1 s. The system in Fig. 17(b) therefore represents a good approximation of the CTDSM to be simulated. The quantizer and feedback DAC can be interpreted as shown in Fig. 17(c), where the output of the quantizer is upsampled by a factor P , interpolated by a P -sample DT filter followed by a CT filter with a rectangular impulse response lasting $1/P$ s. The final simplification is the conversion into a DT system, as shown in Fig. 17(d), where $L_c(s)$ and the zero-order-hold filters are converted into a DT filter $L_1(z)$. The filter output is downsampled by a factor P , quantized, interpolated, and fed back at a rate of P Hz. It is thus seen that while the loop filter operates at a rate P , the quantizer output is updated only once every second. The state matrices of $L_1(z)$ are given [in terms of the state representation of the CT loop filter (A_c , B_c , C_c , and D_c)] by

$$\begin{aligned} A_{d1} &= \exp(P^{-1}A_c) \\ B_{d1} &= A_c^{-1}(A_{d1} - I)B_c \\ C_{d1} &= C_c \\ D_{d1} &= D_c. \end{aligned} \quad (11)$$

Simulating the system in Fig. 17(d) requires only a minor modification in an existing DT simulator (like the simulatedDSM routine in Schreier’s toolbox [16]). P can be chosen so that the internal states of the loop filter are captured with sufficient temporal resolution—in this paper, $P = 32$ was used.

With an appropriate choice of the impulse response of the interpolation filter, the system in Fig. 16(d) can be used to simulate CTDSMs with other feedback DAC pulse shapes. For example,

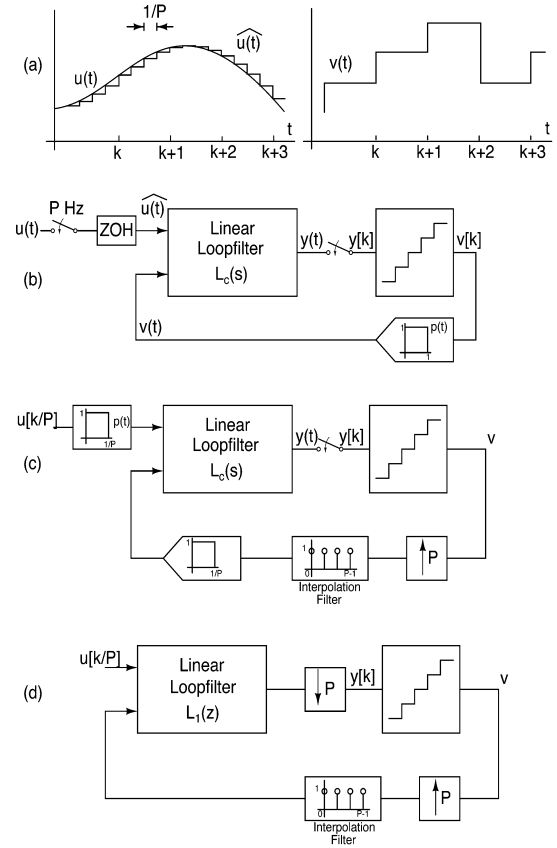


Fig. 17. Simulating a CTDSM in DT. (a) Zero-order-held approximation to the input—sampled at P Hz and the NRZ DAC feedback waveform. (b) Equivalent diagram of the CTDSM with a zero-order-held input. (c) CTDSM with the feedback waveform interpreted as the output of an interpolation filter. (d) CTDSM where the loop filter is simulated at a rate P in DT.

using an impulse train with amplitude 2 and lasting for $P/2$ samples is equivalent to using a RZ DAC. We now consider some practical aspects of the simulation.

1) *Frequency and Impedance Scaling*: Consider a CTDSM operating at a sampling rate f_s and described by

$$\begin{aligned} C\dot{x} + Gx + G_3x^3 &= [F_1 \ F_2][u(t) \ v(t)]^T \\ v[f_s k] &= Q[y[f_s k]] \end{aligned} \quad (12)$$

where $Q[\cdot]$ represents the quantization operation. To avoid round-off/truncation errors that could arise due to the potentially large numbers involved, it is desirable to scale the sampling rate and impedance level of the modulator for simulation purposes. It is straightforward to see that

$$\begin{aligned} \frac{f_s}{k_z} C\dot{x} + \frac{1}{k_z} G_1 x + \frac{1}{k_z} G_3 x^3 \\ = \frac{1}{k_z} [F_{11} \ F_{21}] \times [u(f_s t) \ v(f_s t)]^T \\ v[k] = Q[y[k]] \end{aligned} \quad (13)$$

represents the modulator of (12) operating at 1 Hz, with all impedances scaled by k_z .

Amplitude Scaling in Multibit CTDSMs

Several approaches to describe the operation of an n -level quantizer can be conceived, as described below.

- 1) The quantizer range is fixed ($[-1, 1]$), and the step size is related to the number of levels n as $2/(n-1)$. This is what happens in a practical implementation.
- 2) The quantization step is fixed, and the range is increased depending on the number of levels. This is the approach taken in [16], where the step size is always assumed to be 2, and the n -level quantizer output ranges from $[-(n-1), (n-1)]$.

If the loop filter is linear, approaches 1 and 2 will result in an identical SQNR for the same input (normalized to the quantizer range). If the loop filter is weakly nonlinear [according to (12)], it is easy to show that $\hat{\mathbf{x}} = (n-1)\mathbf{x}$, $\hat{u}(t) = (n-1)u(t)$, $\hat{y}(t) = (n-1)y(t)$, and $\hat{v}(t) = (n-1)v(t)$ satisfy the system

$$\mathbf{C}\dot{\hat{\mathbf{x}}} + \mathbf{G}\hat{\mathbf{x}} + \hat{\mathbf{G}}_3\hat{\mathbf{x}}^3 = [\mathbf{F}_1 \quad \mathbf{F}_2] \begin{bmatrix} \hat{u}(t) & \hat{v}(t) \end{bmatrix}^T$$

$$\hat{v}[f_s k] = (n-1)Q \left[\frac{\hat{y}[f_s k]}{(n-1)} \right] \quad (14)$$

where

$$\hat{\mathbf{G}}_3 = \frac{1}{(n-1)^2} \mathbf{G}_3. \quad (15)$$

From (14) and (15), it is seen that a multibit quantizer with a fixed step size can be simulated by suitably scaling $\mathbf{G}_3$. This was the approach taken in this paper, with the motivation being compatibility with [16].

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